

Polynomial Regression

Polynomial Regression :

⇒ What is P.G.? Instead of using only: x , we can use:

$$x, x^2, x^3, \dots$$

e.g., a quadratic poly. regression model is:

$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2$$

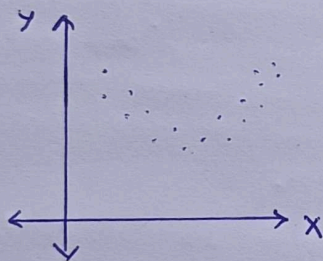
and generally,

$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_d x^d$$

where,

d = degree of the poly.

⇒ It can be applied to nonlinear graphs :



⇒ Why is it called Linear? Bcz the model is linear w.r.t. the parameters β .

⇒ How can PG be treated Multiple L.R. ?
PG can be treated as Multiple LR after transforming the original features into polynomial features.

e.g., suppose our original feature is: x and we create additional features: $x^2, x^3, \dots$

For a degree 2 model:

original x	New feature x^2
1	1
2	4
3	9
4	16

Now, instead of thinking about the model as:

$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2$$

we can think of it as multiple LR with two

input features: $X_1 = x$

$$X_2 = x^2$$

$$\therefore \hat{y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

Bias Variance Trade-off :

⇒ See a model can fail in two ways:

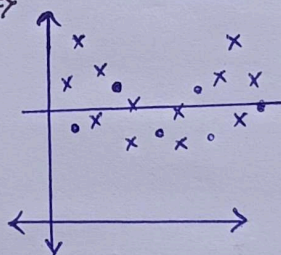
① underfitting → high bias

② overfitting → high variance

⇒ What is Bias? It's the error caused by a model being too simple and making overly strong assumptions about the relationship b/w input and output.

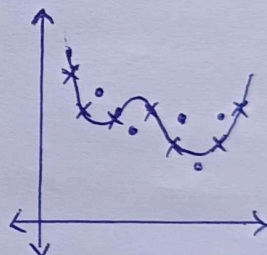
⇒ What is Variance? Variance is about how sensitive the model is to the particular training data it receives.

⇒



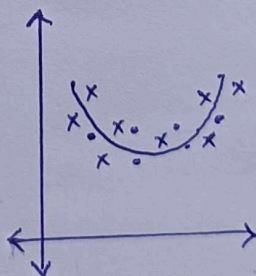
underfitting

- High Bias
- Low Variance



overfitting

- Low Bias
- High Variance



- Low Bias
- Low Variance

* training set
• test set

=> What does "Tradeoff" mean? It means improving one thing can make another thing worse. We need to keep a balance, i.e., bias-variance tradeoff.

=> How to control Bias-variance Tradeoff?

- Regularization
- Cross validation
- Inc. model complexity

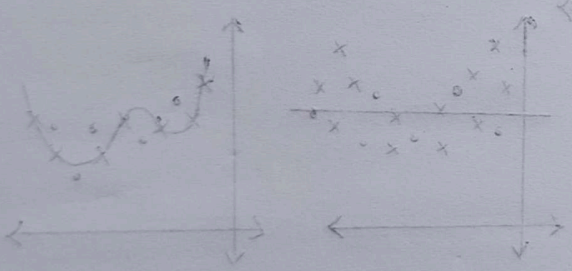
Bias Variance Tradeoff

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- ① underfitting -> high bias
- ② overfitting -> high variance

=> What is bias? It's the error caused by a model being too simple and making overly strong assumptions about the relationship between input and output.

=> What is variance? Variance is how sensitive the model is to the particular training data it receives.

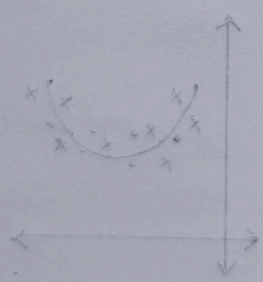


underfitting

- high bias
- low variance

overfitting

- low bias
- high variance



* training set
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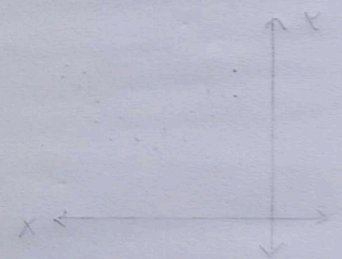
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e.g., suppose an original feature is x and we create additional features: $x^2, x^3, \dots$

For a degree 2 model:

original features	new features
1	1
1	2
2	3
10	10